SMART SK: A WEB BASED PROJECT MONITORING SYSTEM WITH FORECASTING AND PREDICTIVE ANALYSIS FOR SANGGUNIANG KABATAAN OF DISTRICT 5 QUEZON CITY

BONTUYAN, JEFF ALDREICH S.

PROJECT MANAGER

DE GUZMAN, LUIS ALBERT A.

LEAD PROGRAMMER

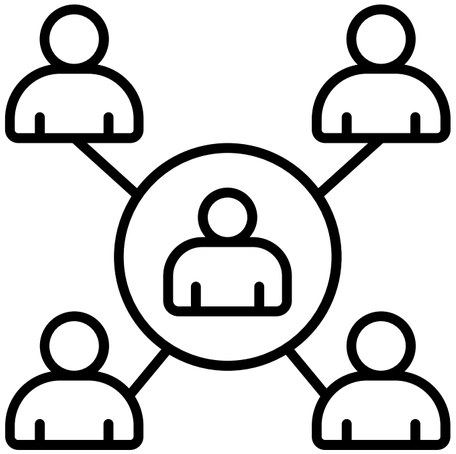
BALAGTAS, REIGN KERSTINE A

SYSTEM ANALYST

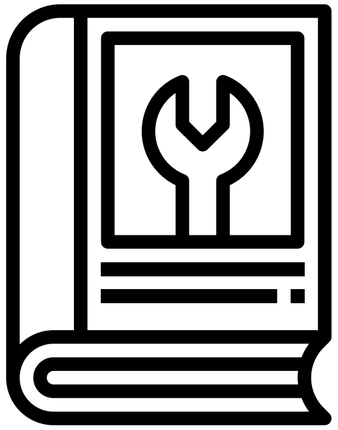
ANDO, YASMIEN M.

QUALITY ASSURANCE

PROJECT CONTEXT

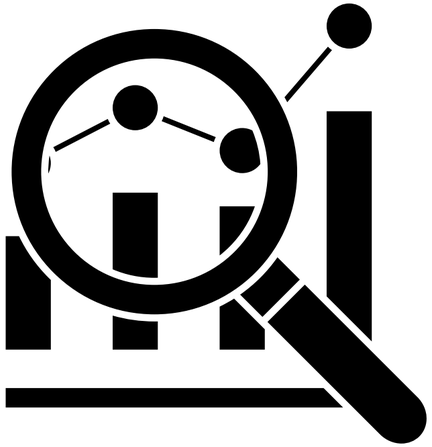


SK plays a key role in youth development.



Current SK processes are manual, inefficient, and unaccountable.

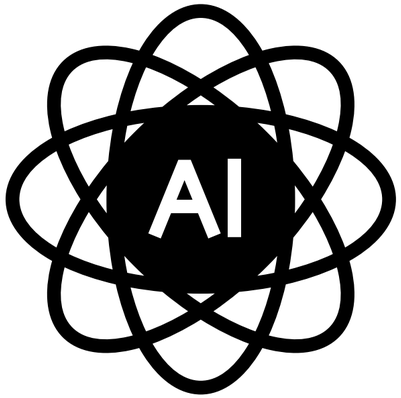
PROJECT CONTEXT



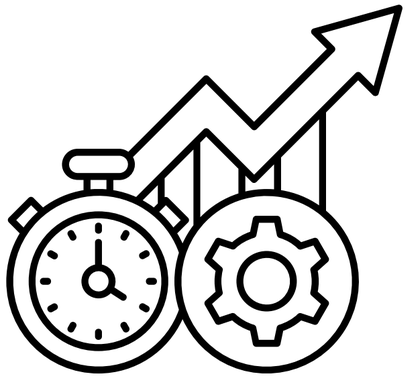
Problems: proposal tracking, project performance.

Proposed solution: Smart SK, a project monitoring system using AI forecasting and predictive analysis.

PURPOSE OF THE PROJECT

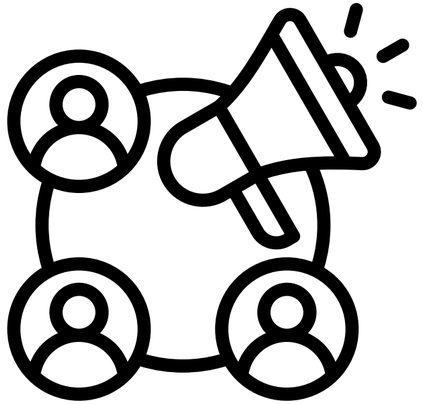


Develop a centralized, AI-powered platform.



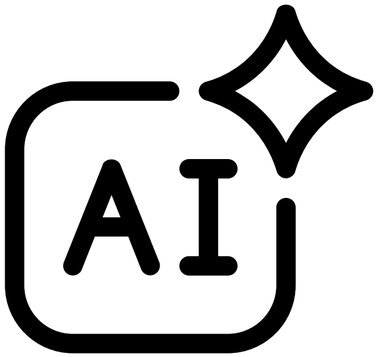
Streamline proposal management, and performance tracking.

Forecast success of proposed projects.



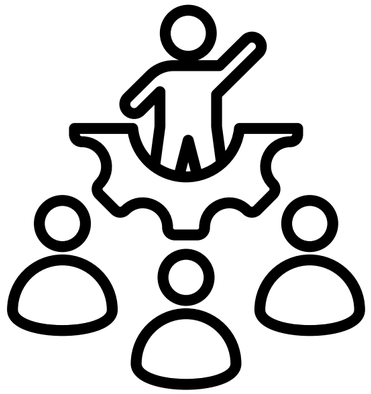
Promote collaboration across multiple barangays.

PROJECT OBJECTIVES



AI-based forecasting for project viability.

Predictive analysis based on historical Data.



Multi-barangay system for district-wide coordination.

SYSTEM FEATURES AND MODULES FRONT-END

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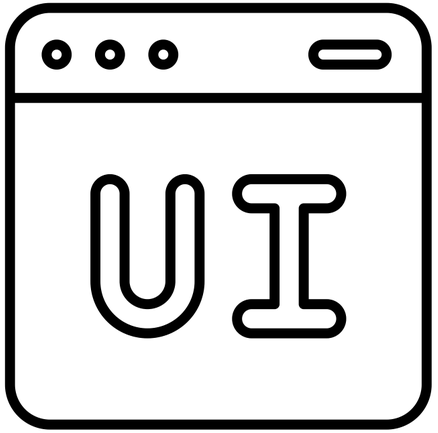
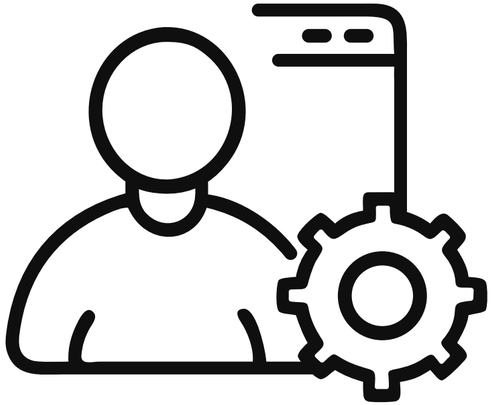
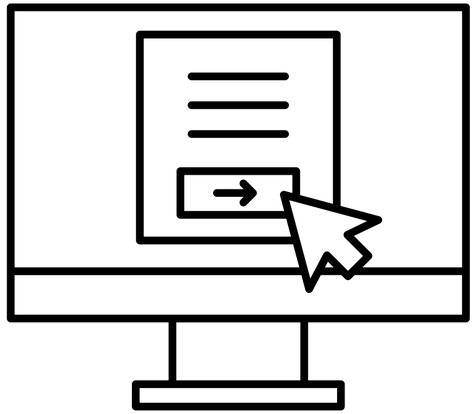


React + TypeScript UI



Dashboard



Project proposal submission and status tracking

SYSTEM FEATURES AND MODULES FRONT-END

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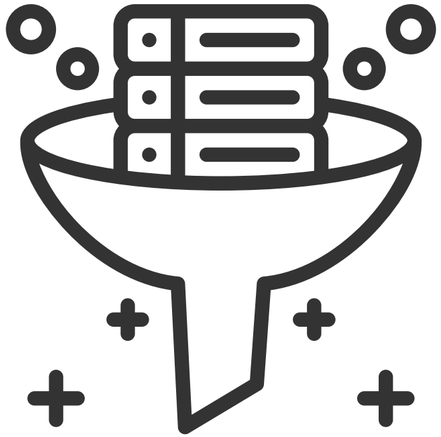
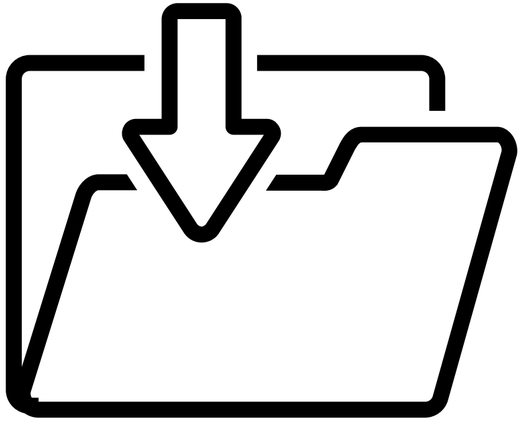
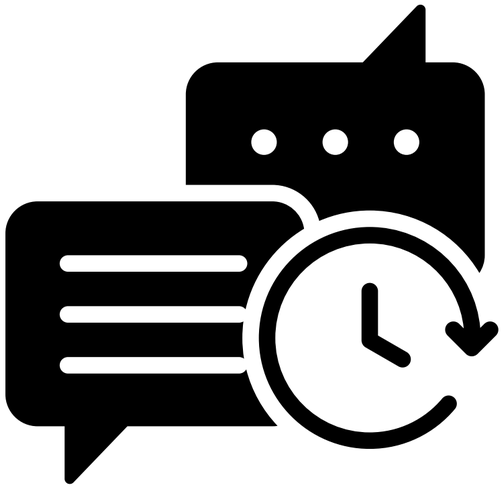


Barangay-filtered view



Archiving of past/rejected projects



Real-time notifications

BACK-END

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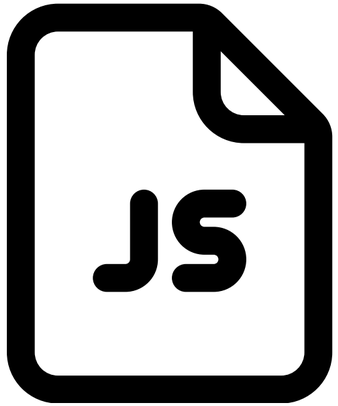
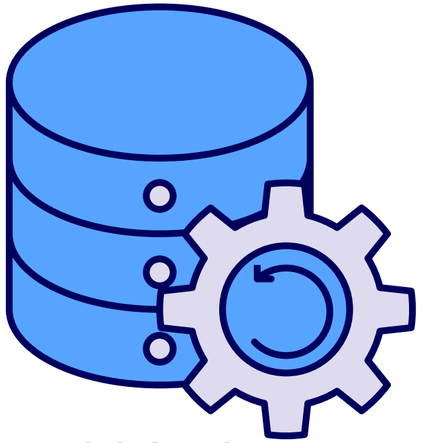


Node.js + Express.js



MSSQL for data storage

AI Forecasting (using Prophet)

BACK-END

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Predictive analysis (Gemini by Google)

Email related logics with Nodemailer

Secure authentication (JWT, UUID4)

AI CAPABILITIES

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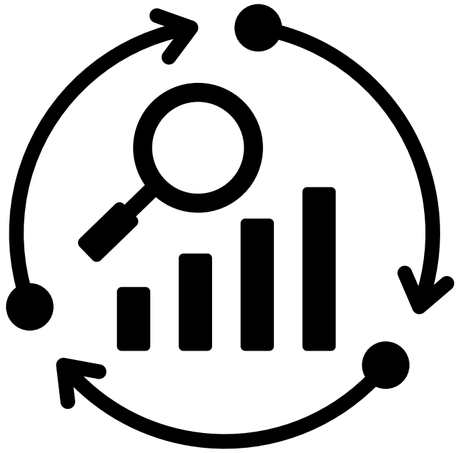

AI Forecasting: Predict number of project implemented using historical data.

Predictive Analysis: Uses historical data & internet sources to provide feedback, suggestions, and/or recommendations for their future projects

TECHNOLOGY STACK

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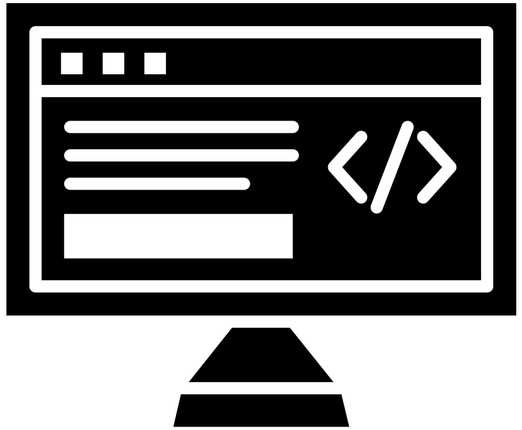
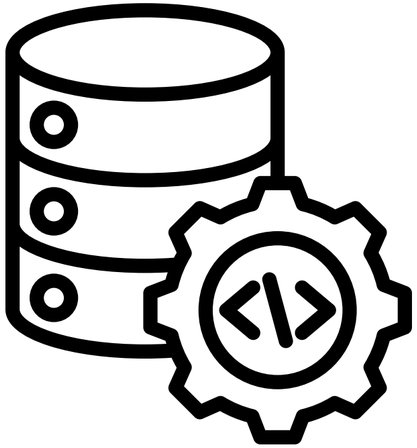


Frontend: React + Vite + TypeScript



Backend: Node.js + Express

TECHNOLOGY STACK

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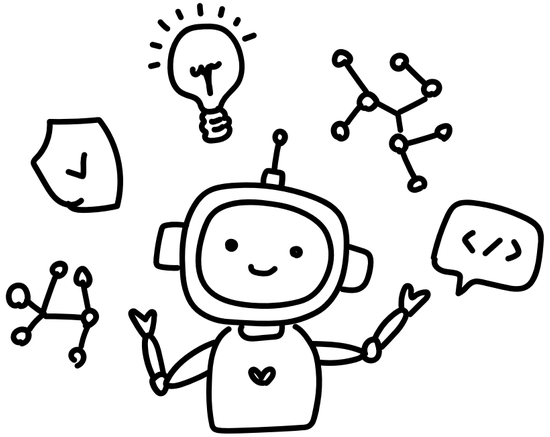

AI Tools: Python, Prophet (time-series forecasting), Gemini (predictive LLM)

Database: Microsoft SQL Server

METHODOLOGY

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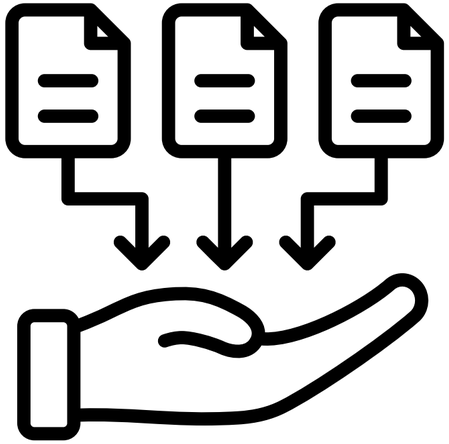
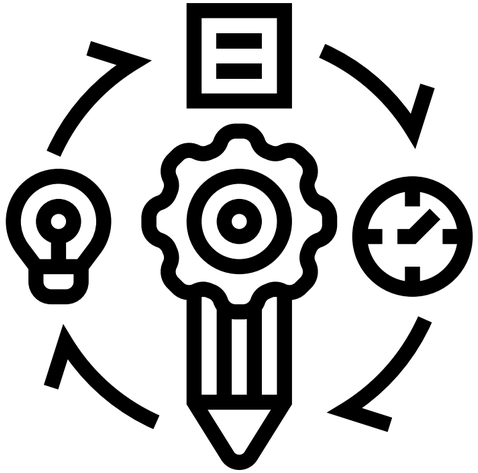
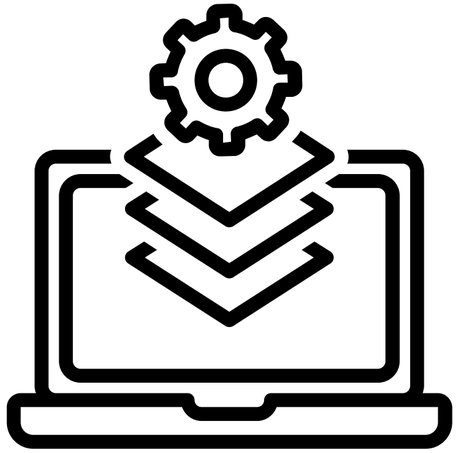


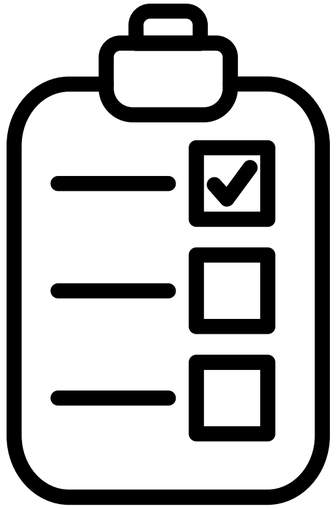
Data Gathering: Interviews, observations, document reviews



Design & Planning: Notion, Figma, whiteboards



Development: Full-stack coding with modular APIs



Deployment Prep: Security, performance, scalability

LIMITATIONS

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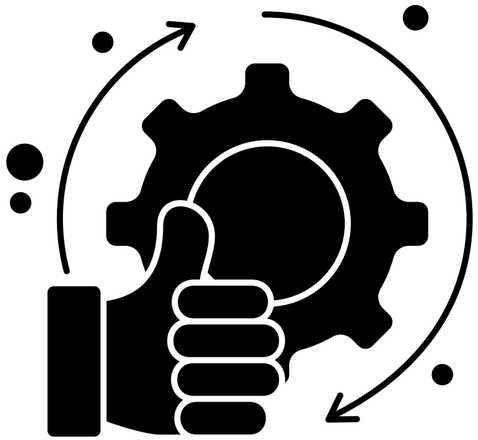
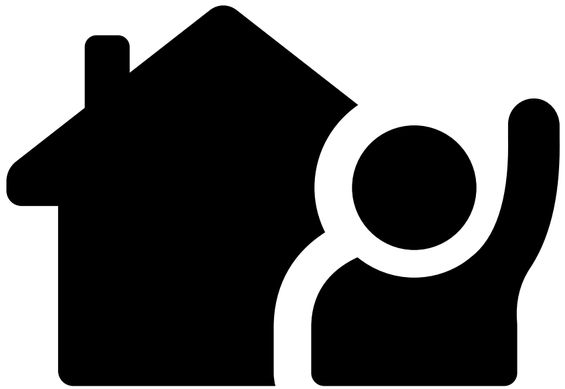


Accuracy of AI depends on data quality and quantity.

Internet dependency may hinder accessibility.



Digital literacy gap

LIMITATIONS

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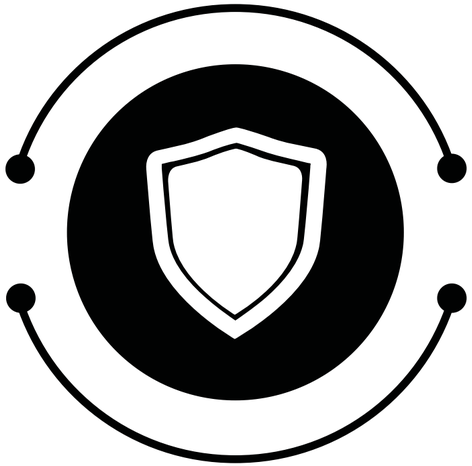
Initial manual data entry is needed.



Data privacy concerns must be addressed.

Lack of Historical Data.

REVIEW OF RELATED LITERATURE AND SYSTEMS

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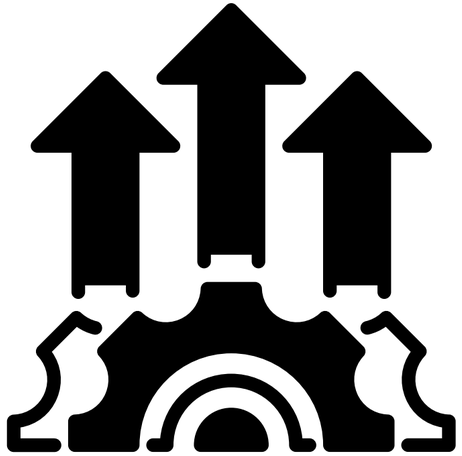
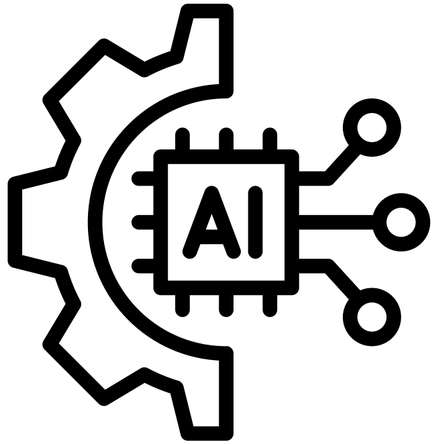


AI improves efficiency, transparency, and governance.



AI in education, health, transport, and project management shows success in forecasting and automation.

REVIEW OF RELATED LITERATURE AND SYSTEMS

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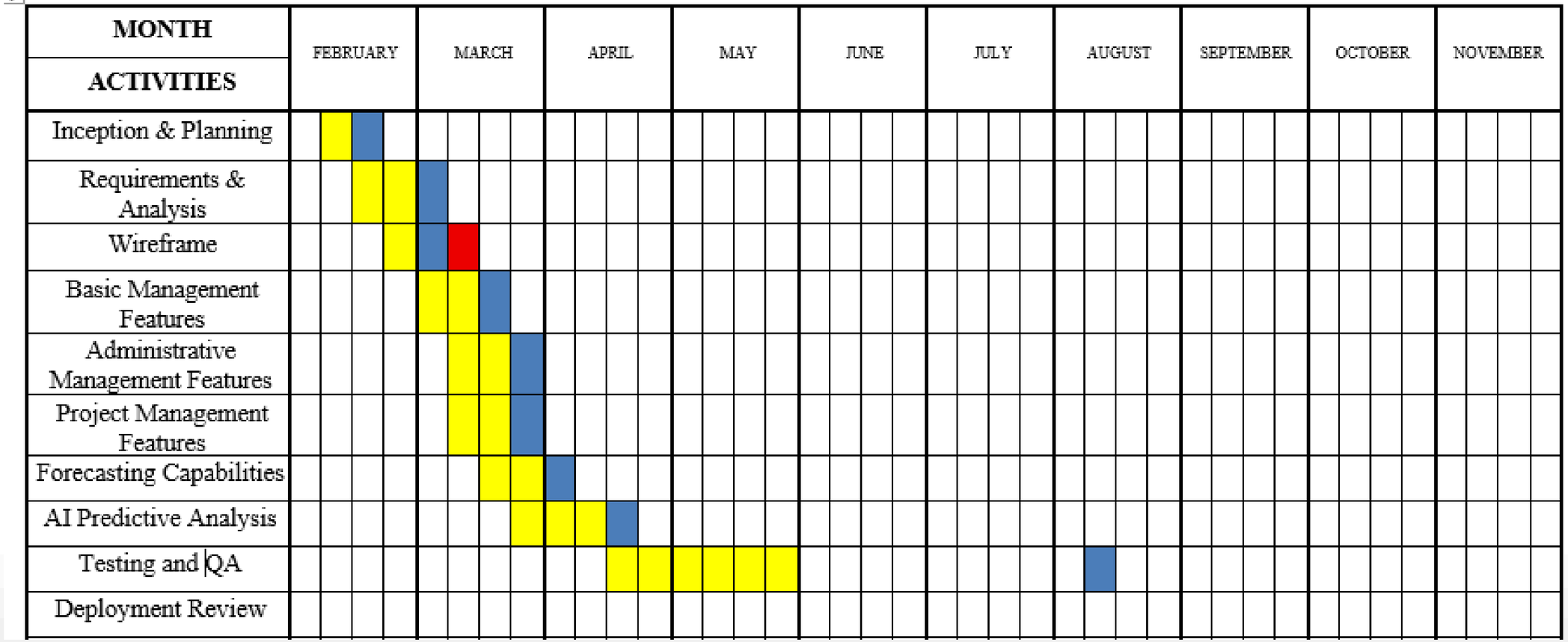
Local systems (Barangay MIS, SK IMS) prove the viability of digitization.

Forecasting models (Prophet) effective for time-series and project tracking.

GANTT CHART

Gantt Chart of Activities

Legend: Blue - End, Yellow – Start & Red - Delay



HARDWARE & SOFTWARE RESOURCES

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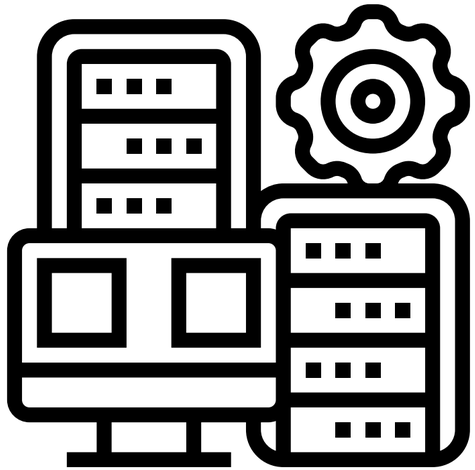
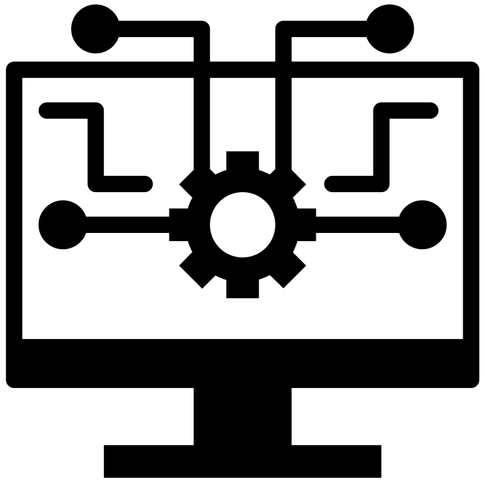


HARDWARE: Windows 11
Intel i5, 24GB RAM



SOFTWARE: VS Code, TypeScript, Node.js, Python, MSSQL, React, Express.js, Gemini, Prophet

CONCLUSION

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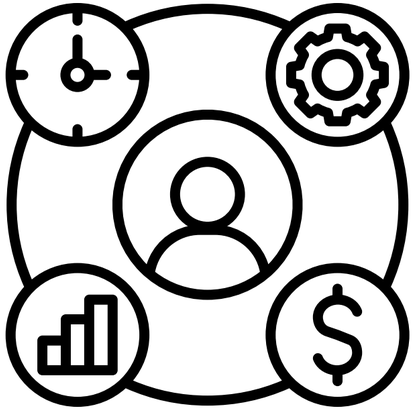
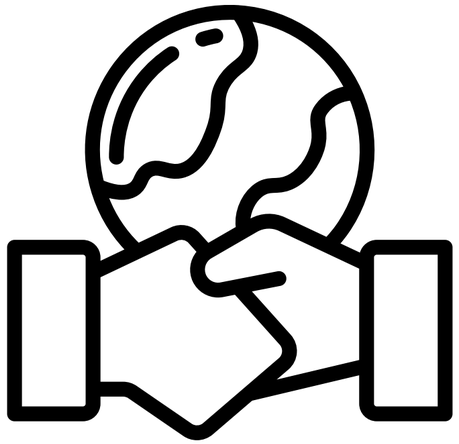
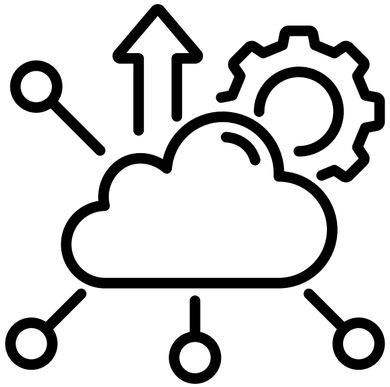


Smart SK transforms SK project management with AI.

Promotes transparency and youth involvement.



Fosters collaborative governance across barangays.



Deployment Prep: Security, performance, scalability

CONCLUSION

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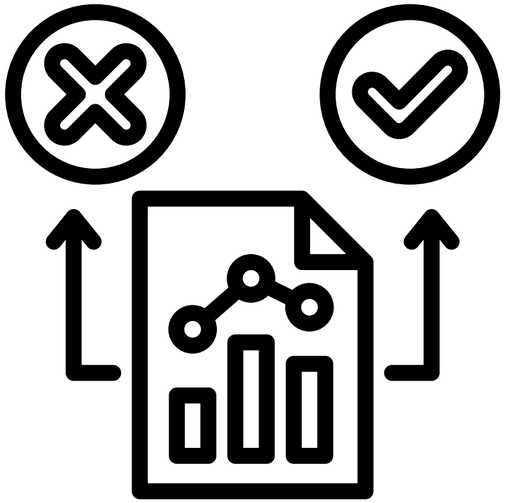

Enables data-driven decisions and resource optimization.

Long-term impact: Sustainable youth development programs.